

Software Verification and Validation (BITP3253)

Project Description

1. General Information

- **Course Name:** Software Verification and Validation
- **Course Code:** BITP3253
- **Team Structure:** 6 Members per Team (from similar Lab Group)
- **Submission:** Week 13
- **System Type:** Mobile & Web-Based System

2. Project Objective

The goal of this project is to apply advanced software verification and validation techniques to a multi-platform system. The team is responsible for ensuring the system's reliability, performance, and usability across both web and mobile platforms.

3. Scope of Testing

The project will focus on three critical testing domains:

1. **Interface Testing:** Evaluating the interaction between the system's components and ensuring the UI/UX is consistent across platforms. The test interfaces include user login interface, query user profile interface, update user profile interface and other interfaces. You need to design test cases, execute interface tests and record the process according to the test objectives and test content.
2. **Performance Testing:** Performance testing (JMeter) Apache JMeter and PerfMon Plugin tools (The PerfMon Plugin used in conjunction with JMeter to monitor server resources during test execution.) Assessing system responsiveness, stability, and scalability under load. You are required to carry out performance testing activities on your system, identifying program performance issues that were not discovered by the developers, as well as locating API interfaces with slow response times. Based on the testing objectives and content, you should write or record performance testing scripts, identify performance defects, and produce a test report.
3. **Automated UI and functional testing:** (Selenium WebDriver 4.x + pytest 7.x + Allure Report 2.x). Design and implement automated UI tests using Selenium WebDriver + pytest. Structure all tests using the Page Object Model (POM) pattern. Generate and submit a structured Allure Report
4. **Functional Testing:** (mobile app) (Android Emulator and Appium). Verifying that all features of the selected application operate according to functional requirements. Test scripts shall

be written in Python and developed with PyCharm. Test data for the DDT Testing. Choose any mobile app, which may contain defects. As a software tester, your task is to analyze the requirements in the app, conduct automated testing based on test cases (scenarios) that you plan (automated), write automated test scripts, execute automated tests, and output an automated test report.

4. Technical Tools & Infrastructure

The project requires the use of five specialized tools. Four are compulsory, and one is selected by the team to best fit the project's needs.

4.1 Compulsory Tools

- **JMeter:** For performance and load testing.
- **Postman:** For API and integration testing.
- **Selenium:** For automated web-based functional testing.
- **Appium:** For automated mobile-based functional testing.

4.2 Elective Tool

- **[Student's Choice]:** The team will select one additional tool (e.g., Katalon Studio, SonarQube, or Cypress) to enhance the verification process.
- **[Student's Choice]:** If the system leverages **Firebase**, you can get more significant advantages.

5. Deliverables & Submission

Final results must be submitted through **Ulearn**, rubric will also be given later.

The submission package must include:

1. **Project Report:** A comprehensive document detailing the three critical testing domains, test cases for each domain, execution results, and bug reports.
2. **Demonstration Video:** A video recording showcasing the automated test execution and a summary of the findings. During the recording, showing your face is compulsory.